HoloKai DeBERTa GlobalPointer comparison with the midterm GLiNER model

The DeBERTa GlobalPointer model achieved mean default-threshold V4 F1 90.09% across 3 seeds, a difference of +1.84 percentage points from the saved original GLiNER model. The comparison uses the original midterm annotations, not Option B.

Overall results

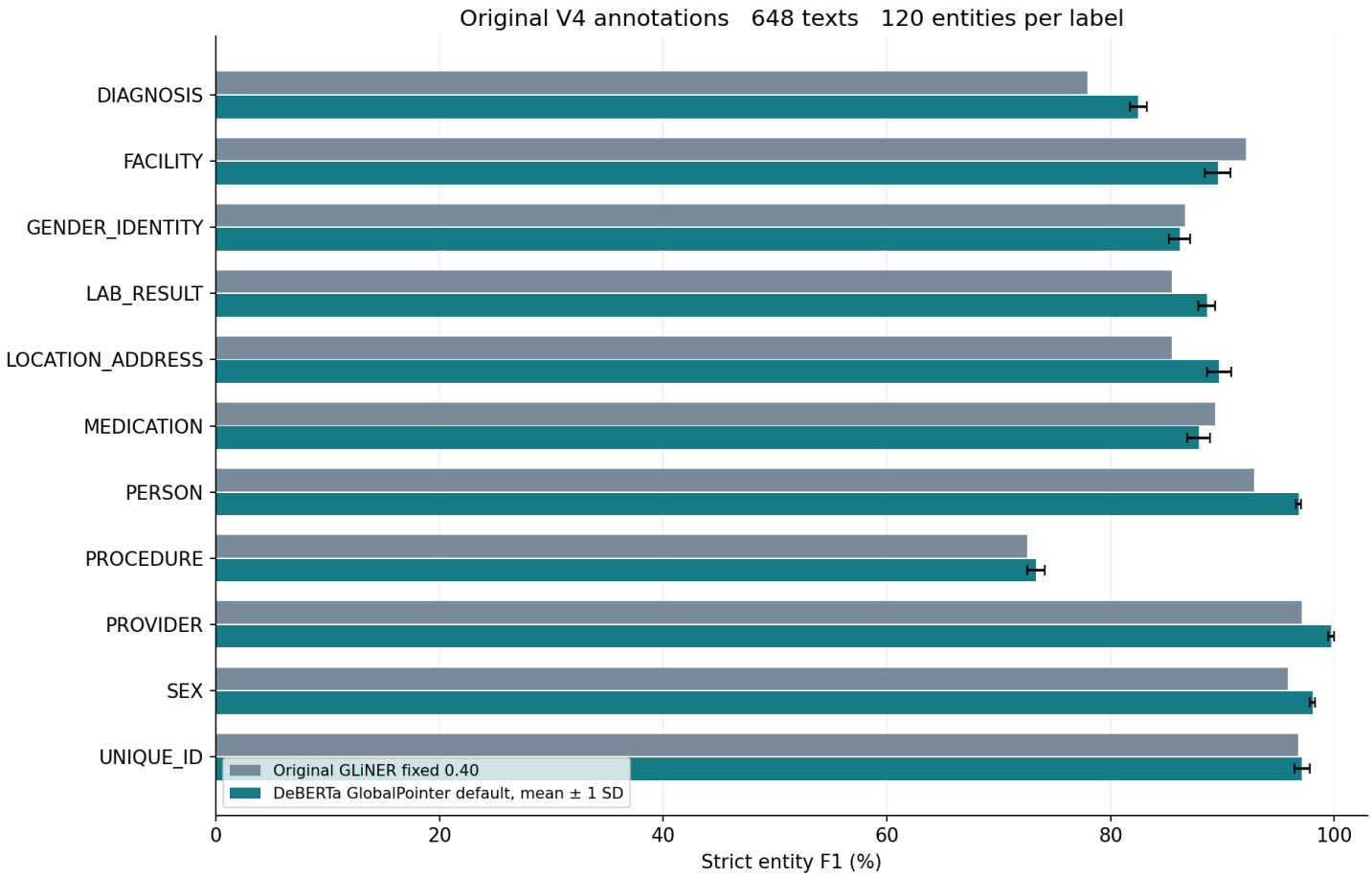
Model	Seed	Precision %	Recall %	F1 %	TP	FP	FN
Original GLiNER pretrained fixed 0.40	pretrained	47.62	60.61	53.33	800	880	520
Original GLiNER fixed 0.40	one saved model	86.15	90.45	88.25	1194	192	126
Original GLiNER validation per label	one saved model	87.16	89.47	88.30	1181	174	139
DeBERTa GlobalPointer default_logit_0	42	92.20	87.73	89.91	1158	98	162
DeBERTa GlobalPointer validation_global	42	92.20	87.73	89.91	1158	98	162
DeBERTa GlobalPointer validation_per_label	42	92.18	86.67	89.34	1144	97	176
DeBERTa GlobalPointer default_logit_0	7	91.48	88.71	90.08	1171	109	149
DeBERTa GlobalPointer validation_global	7	93.41	86.97	90.07	1148	81	172
DeBERTa GlobalPointer validation_per_label	7	91.90	87.73	89.77	1158	102	162
DeBERTa GlobalPointer default_logit_0	99	92.39	88.26	90.28	1165	96	155
DeBERTa GlobalPointer validation_global	99	92.39	88.26	90.28	1165	96	155
DeBERTa GlobalPointer validation_per_label	99	93.62	86.67	90.01	1144	78	176

Variation across random seeds

New model threshold	F1 mean %	SD pp	Seeds
default_logit_0	90.09	0.18	3
validation_global	90.09	0.18	3
validation_per_label	89.71	0.34	3

SD is sample standard deviation across new-model training seeds. The old model is one saved checkpoint; its seed variability was not rerun. Default thresholds differ by model: GLiNER probability 0.40 and GlobalPointer raw logit 0. They are not equivalent confidence scales.

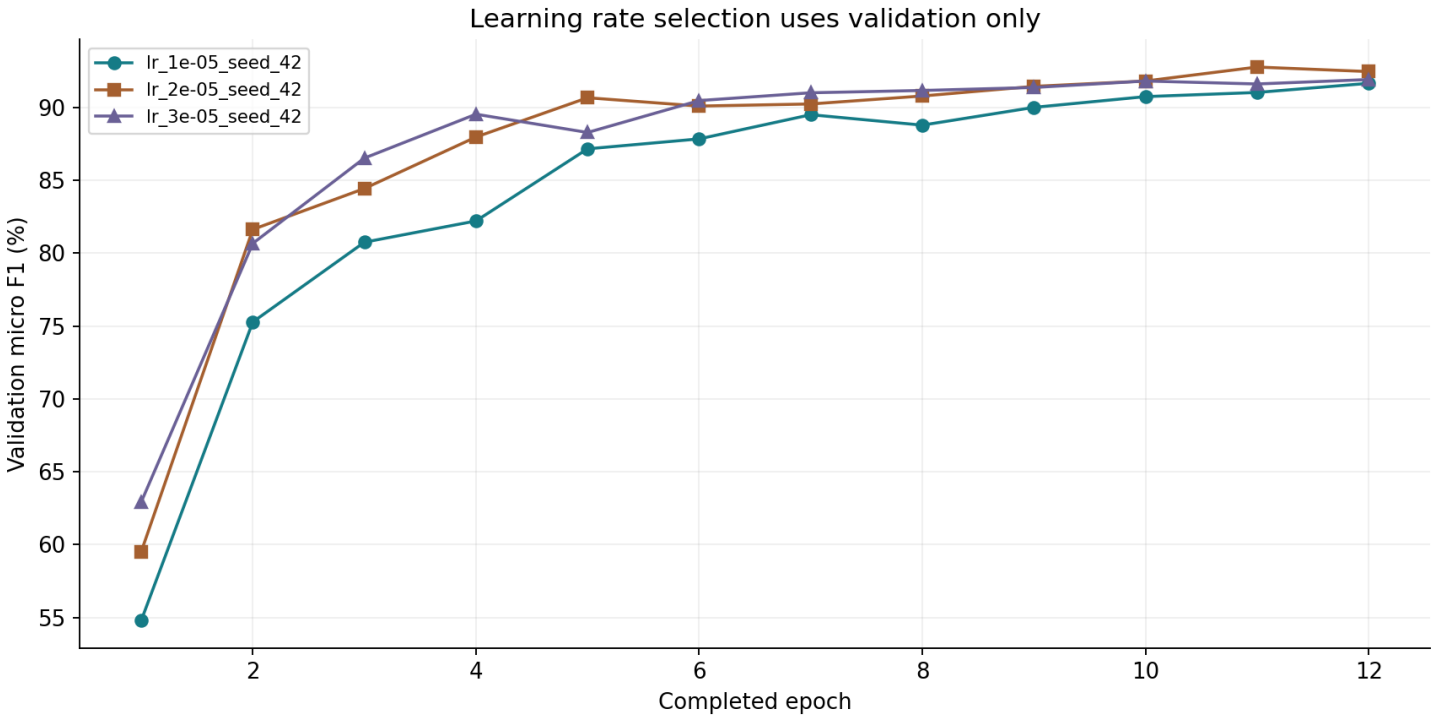
Results by entity type



Label	Gold	Old fixed F1 %	New default mean F1 %	Change pp	Old calibrated F1 %	New calibrated mean F1 %	Change pp

DIAGNOSIS	120	77.91	82.48	+4.57	76.74	82.69	+5.95
FACILITY	120	92.12	89.59	-2.52	92.12	89.56	-2.55
GENDER_IDENTITY	120	86.67	86.17	-0.49	88.21	84.99	-3.22
LAB_RESULT	120	85.49	88.60	+3.11	84.52	88.23	+3.71
LOCATION_ADDRESS	120	85.48	89.71	+4.23	85.47	88.77	+3.30
MEDICATION	120	89.36	87.88	-1.48	89.74	85.56	-4.19
PERSON	120	92.86	96.81	+3.95	92.86	96.69	+3.83
PROCEDURE	120	72.52	73.36	+0.84	72.87	74.12	+1.26
PROVIDER	120	97.12	99.72	+2.60	97.12	99.58	+2.46
SEX	120	95.87	98.06	+2.19	95.87	97.47	+1.60
UNIQUE_ID	120	96.75	97.12	+0.37	97.14	97.25	+0.10

Validation experiments



The learning rate and checkpoint were selected using validation micro F1 at raw logit zero.

Earliest epoch wins checkpoint ties; smaller learning rate wins sweep ties. Each selected seed uses its own validation-selected checkpoint and thresholds. No seed was selected by V4 score.

Data and scoring

Train: 1509 texts and 3947 entities. Validation: 171 texts and 453 entities. V4: 648 texts and 1320 entities, 120 per category. Strict inclusive original-token boundaries and labels are matched within each sample, using micro averaging. Duplicate predictions count as extra false positives. Nested decoding is permitted, but these annotations contain no nested entities, so nested-subset performance is not estimable.

Interpretation limits

This is a controlled data and scoring comparison, not a matched tuning-budget or training-objective experiment. The old GLiNER checkpoint was selected in a larger historical search; the new model uses a three-learning-rate sweep. The presentation reports earlier V4 evaluations during development, so V4 is an established reused project benchmark rather than a newly untouched external test. The new experiment freezes its choices before V4 evaluation. Direct runtime comparison is unavailable because the old saved metrics do not record comparable timing; only new-model timings are reported.

Sources and files

User-selected midterm presentation. Exact old scores were read from the original final_model outputs, and its fixed-threshold scores were recomputed from saved predictions. Raw tables: overall_comparison.csv, per_label_comparison.csv, seed_summary.csv and learning_rate_history.csv. Error files contain original text and should remain with the project data.

Frozen protocol SHA256:
56d939f8b507c57e99fe3066b813a663cb5bc501478976c04c68ea01b0507639
V4 SHA256:

ba40da69601298e626a3e748640bf1787c541b56a63871161e4e2044bda39be6